
JPA FUNDING STRATEGIES

Vance Russell

3point.xyz, Forest Business Alliance

11th Jun, 2025

created in  Curvenote

Abstract

Abstract: An analysis and strategy for securing sustainable funding for joint powers authorities (JPAs) as woody feedstock aggregators is introduced. JPA entities are proposed to create to improve forest product supply chain bottlenecks. An estimated \$1.5 mn is needed to start up and run the Northeast California JPA over three years. Startup funding for each of the pilots is being funded by the California Governor's Office for Land Use and Climate Innovation. Long-term, sustainable funding for each JPA will be challenging, especially in rural areas without the tax base to support wildfire mitigation authorities, such as Marin or JPAs, which help facilitate the utilization of wood from thinning projects. A portfolio approach to securing public and private funding, as well as self-sustaining revenue sources, is among several recommendations.

Keywords b, i, o, m, a, s, ,, , f, n, c, e, d, t, k, r, h, l, u, g, j, p, w, y

1 JPA Funding Strategies

Joint Powers Authorities are proposed as a solution for feedstock aggregation in California to help broker agreements with forest landowners and managers and create long-term supply contracts to increase investor support for wood products businesses. Funding these entities presents its challenges, given the difficulties of securing grants and private investments. Creating a diversified funding portfolio is crucial, especially for new entities, and suggestions for achieving this are provided.

1.1 Takeaways

1. **Portfolio strategy.** Be strategic about what funds you utilize for different projects and programs. It may be best to look to foundations and business support grants for organizational startup or administrative costs. Still, implementation grants for field projects and private or bank loans for scaling once a feedstock aggregation entity is established.
2. **Temporal ramping.** Revenue sources outside the realm of taxes and grants may be difficult to create and execute with risk-averse agencies and organizations. Try funding with traditional resources, such as grants, first, and then gradually introduce new resources to diversify your funding portfolio and help establish a new organization's financial stability over time.
3. **Grant dependence & giving.** In a similar vein, just because grants are available doesn't mean that will be the case indefinitely. As soon as you secure a grant or two, think about how to leverage it with other funding sources and work tirelessly to create more dependable and steady income sources than grants. It may seem daunting, but remember that the single largest source of charitable giving in the United States comes from individuals, not foundations, corporations, or grant-making agencies. The single largest source is also the least restricted type of funding, so it is critical for covering operational expenses (which are often underfunded by grants). Develop a strategy to cultivate local support through individual donations.

1.2 Background

The large wildfire seasons of the past five years have been catastrophic for forests and communities. However, according to ?, they are not unprecedented in recent California history and are usually associated with periods of drought, e.g., the 1920s and 2010s. ? found that during the past century, tree densities increased up to sevenfold while average tree size decreased 50% in the Sierra Nevada. ? led a meta analysis showing a fire deficit and widespread alteration of ecological structure and function across seasonally dry forests of western North America. A chronic level of stress is created by high competition across tree stands, resulting in reduced resilience to drought, disease, fire, and climate change. ? advocates for an aggressive approach to treating fire-suppressed stands using an ecologically based approach that reduces the total number of trees/acre while maintaining stand heterogeneity.

1.3 Concerns

A significant cause for concern regarding the protection of old-growth forests in the United States is their decline from historical levels due to logging and development. ? highlight the fact that a large proportion of the remaining mature and old-growth forests on federal lands are still vulnerable to logging. The loss of these forests would not only diminish biodiversity but also release substantial amounts of stored carbon into the atmosphere.

Further compounding these concerns is the lack of a coordinated and effective national policy for old-growth conservation. ? point to a history of policy debates, such as the National Old-Growth Amendment, which has failed to provide lasting protection. The authors advocate for a more comprehensive approach that extends beyond simple harvest restrictions to encompass landscape-level planning, the establishment of reserves, the protection of climate refugia, and the setting of specific goals for the recovery of species that depend on these unique habitats.

But let's be very clear: thinning dense stands of fire-prone forests is completely different from logging old growth, and ecological-based thinning is not an excuse to log when maintaining stand heterogeneity and reducing fire risk are the treatment goals. Yet, misinformation about wildfire and forest treatments, similar to climate misinformation, persists (?).

Managers implementing forest health treatments should adopt a tailored approach to increase forest resilience, mitigate fire risk, and bring stands within a natural range of variation.¹ Treating mixed conifer stands on the western slope of the Sierra Nevada requires different treatments than mesic forests in the Pacific Northwest. Managers should take steps to reduce fire risk while still considering the negative impacts on biodiversity when planning any fire mitigation project.

1.4 Feedstock Aggregation

Thinned forests create a lot of biomass, and much of that biomass from forest health projects end up in burn piles or log decks and may stay in these locations indefinitely. Not only does this negate fire mitigation efforts, as burn piles and decks can promote or worsen fires, but the carbon from wood is also at risk of not being sequestered.

Processing biomass from thinning is challenging; ? found that the cost of forest restoration far exceeds current market prices for biomass. ? point out that supply guarantees, industry presence, transportation, and the value of the biomass are limiting to biomass utilization, whereas agency staffing, budgets, compliance, and partnership aggravated use rather than impeding progress. SIG recently completed a burn pile inventory across California, showing massive tonnages of piles scattered through national forests and other lands. Yet burning those piles is likely more expensive than transporting them to a facility for processing (?). Transportation subsidies to move this feedstock to central locations or nearby facilities could be a crucial missing component in addressing the wildfire problem across the western United States.

Transporting the feedstock to a central location, accessible and central to processors, biomass facilities, and wood product businesses, would help move the biomass out of the woods, mitigating fire risk while also facilitating the centralization of long-term feedstock contracts with landowners and managing agencies. For instance, [Regenerative Forest Solutions](#) identified approximately half (242,365 acres) of Sonoma County's forested acres as feasible for treatments (Figure ??). Berry Sawmill was identified as an ideal location for a wood products campus and

¹See (?).

[illegible]

1.5 Joint Powers Authority

Joint Powers Authorities are legally created entities that allow two or more public agencies to jointly exercise common powers (?). A joint powers authority shares powers common to the member agencies, and those powers are outlined in the joint powers agreement. Joint powers are exercised when the public officials of greater than two agencies agree to create another legal entity or establish a joint approach to work on a common problem, fund a project, or act as a representative body for a specific activity. Examples of areas where JPAs are used commonly include groundwater management, road construction, and habitat conservation (?).

Biomass insurance from Ecostrat? How does this generate revenue for a JPA? <https://ecostrat.com/feedstock-supply-insurance>

In establishing the JPA, partners have been clear about its role: to act as a critical entity for wood utilization without competing for funds with other organizations, such as Resource Conservation Districts (RCDs), sawmills, or licensed timber operators (LTOs). This approach requires a realistic revenue assessment and a plan for the initial five years of operation, acknowledging that creating a stably funded JPA will be challenging given the wide fluctuations in private and public funding. But how can such an entity be funded in rural counties with low tax bases that cannot support a sales tax to develop sustainable revenue for a new entity?²

2 Funding Options

The abundance of state and federal funding over the past five years has meant many agencies and organizations depend on grant funding for implementing restoration and infrastructure projects. Trump administration cuts in 2025 demonstrate the vulnerability of relying too heavily on grant funding. Many nonprofits are familiar with the cyclic nature of many funding sources.

2.1 Traditional

The Northeast California CAL FRAME project recognized the need for grant funding to initiate a JPA and acknowledged the need for additional funding sources that can sustainably support a JPA over time. These are described below and summarized in (Table 1).

1. **Endowment.** Create an endowment from the OPR funding tranche (plus other contributions) and invest it conservatively. Interest or gains from the principal would partially fund a JPA.
2. **Contributions.** Although the McConnell Foundation is an eventual, not immediate donor, casting a wider net at the state or Western Region level for foundations, corporations, and individuals that could give to a JPA is warranted. We suggest that rather than considering the contributions as grants to support a JPA, the Pilot Project Team should consider creating a campaign with a specific fundraising amount to start. This approach will make it easier to approach each donor with a specific amount to ask, involve donors in the overall campaign, and make smaller contributions from individuals more meaningful and achievable. We also recommend that initial contributions go to endowment creation.
3. **Federal and State Grants.** It is unlikely that a JPA will receive operational funding from state and federal grants, but any proposals it leads or participates in could charge overhead (~10%) plus directly bill salaries to cover some operating costs. Negotiating a higher indirect, or NICRA, rate to increase this value through a grant could be another long-term strategy to boost operating cost revenues.
4. **Member Contributions.** RCD and other feedstock aggregation members pay \$5k/yr to participate in the program. This would be unrestricted funding for a JPA.
5. **Fee for Service.** Western Shasta RCD, for example, works with nonindustrial forest owners and provides services for forest management plans with small forested LOs (The State funds the development of FMPs, but landowners cannot secure funds to implement the plans). JPA could offer a similar service. However, it must ensure it does not compete with the region's RCDs.
6. **Sort yards.** Managing a sort yard for aggregated feedstock would be a strategic revenue source. This would require a substantial investment in equipment and a suitable site. See the budget for a cost breakdown. A feasibility study would help understand the total costs, potential revenue, and risk reduction associated with the yard.

2.2 Temporal Adoption

In general, the feasible options identified are traditional and similar to funding approaches for an RCD or nonprofit.

²JPA's similar to the Marin Wildfire Prevention Authority, funded through sales tax increase, aren't possible in northeastern California where a low tax base and few people cannot generate the same income as a more densely populated county such as Marin.

Table 1: Analysis of funding types appropriate to aggregation JPAs. Timing roughly refers to the amount of time required to generate income. Difficulty is a qualitative scale ranging from 1 (easiest to secure) to 5 (most difficult).

Type	Pluses	Minuses	Timing	Difficulty
Endowment	Steady income from principal	Need significant amounts to generate income	May take a long time to build principal	5
Individual Contributions	Unrestricted funding	Much time to manage individual contributions	Instant	1
Public Grants	Access to large amounts of money	Reimbursable, administration costs can be high, stiff competition, cyclic	Public grants may take up to 1 year to contracting from proposal submittal	3
Private Grants	Often include unrestricted funding	Dependent on relationships with program officers, board members	Lengthy time for relationship building	2
Fee-for-Service	Steady income based on capacity and experience for services	Staffing, unevenness of demand	Long time to develop and market services	3
Sort Yard	Steady income, demand for wood high	Transportation costs, management, long-term supply	Permitting and capacity building may take a long time	4

1. New finance options, such as environmental impact bonds, could be introduced over a longer period and trialed at a small scale, then scale up when successful. These could also include other financial mechanisms, such as revolving loan funds (offering low-interest loans), public-private partnerships, or bonds.
2. Other revenue-generating options could be tested in the same manner to ensure their effectiveness and introduce them to skeptical participants, financial managers, or government entities that are not used to working with alternative finance options.

2.3 Collaborative Finance

Collaborative finance is a conservation finance strategy that involves cooperative interaction between individual project developers, stakeholders, and finance providers. This process may or may not include traditional financial institutions.³ The term can be broadened to include finance developed through the fair and equitable participation of stakeholders in a region, landscape, or watershed, addressing natural resource and infrastructure management needs, and utilizing multiple forms of funding, from public grants to private investment. Finance approaches may include outcomes-based finance models such as environmental impact bonds.⁴

New Funding Options

New Finance Options Conservation finance options include environmental impact bonds, such as Blue Forest's Forest Resilience Bond, which leverages private investment from impact investors to support forest health projects. Repayment to investors is supported over time by beneficiaries based on the savings and benefits from reduced wildfire risks and improved ecosystem services. A JPA could utilize environmental

³See [collaborative finance](#) for more information.

⁴For a deeper discussion of collaborative finance approaches to financing water infrastructure and fire mitigation, see [Because It's Worth It](#) and [Conservation Finance: Towards a new model for landscape restoration](#).

impact bonds on a trial basis at a small scale and then expand over an extended period. Identifying potential payor entities interested in covering the avoided costs of wildfire through proactive measures, such as thinning and prescribed fire, is worth pursuing immediately or as soon as an executive director is hired.

We do not recommend relying solely on impact bonds, as they take a lengthy time to develop and rely on a payor to provide avoided cost funding. However, a bond could be a valuable complementary funding resource to other secured revenues. Other 'newer' finance options could include the following:

Carbon Markets Carbon markets provide an opportunity to secure funding for on-the-ground forest management activities, including thinning, pruning, mastication, mechanical treatment, and prescribed burning. Revenue realized through the carbon market could help a JPA to treat more forest land than otherwise possible and generate additional biomass that could be put under supply contracts to support existing and emerging infrastructure. Market prices for carbon credits vary depending on a given project's size and location, treatment type, and the carbon market or registry used. Carbon credits can be realized for projects of any size and located on federal, state, and private lands. Refer to the Avoided Wildfire Emissions Protocol section for an alternative approach that is easier to implement than carbon sequestration credits or markets, as the avoided costs are already calculated, and proponents do not need to seek an entity to purchase the credits.

The National Forest Foundation implemented several voluntary carbon projects in California and the West following wildfires. Funded by corporate donors, the projects created carbon based on growth trajectories of planted trees. The credits were registered with the American Carbon Registry but were immediately retired upon creation, as trading carbon on public lands is prohibited.

Avoided Wildfire Emissions The Spatial Informatics Group and Element Markets developed a forecast methodology under the Climate Forward program to recognize the climate benefits associated with fuel treatment activities that reduce the risk of catastrophic forest fires and their associated emissions. Known as the Avoided Wildfire Emissions Forecast Methodology, the Climate Action Reserve protocol could provide complementary funding for thinning and prescribed fire projects to grants and private investments. The Protocol differs from carbon offsets in that forecasted mitigation units (FMUs) are issued for forecasted greenhouse gas reductions or removals. FMUs are used to mitigate anticipated future emissions, such as wildfires.

Embodied Carbon Although it may be several years from implementation, developing and marketing low-carbon building materials is another new opportunity. Buildings are a significant source of greenhouse gas emissions, making building decarbonization a California state priority. Embodied carbon is the lifecycle of greenhouse gas emissions from creating, transporting, and disposing of building materials (Carbon Leadership Forum, 2022). In other words, embodied carbon is any building's carbon footprint contained in its building materials. It differs from operational carbon, the carbon produced by a building's energy, heat, and lighting. The California Air Resources Board is developing a comprehensive strategy for embodied carbon and is seeking comments following the signing of ABs 2446 and 43.

Revolving Loan Funds Pooled funding sources such as impact bonds or revolving loan funds can help make a greater amount of funds available to more projects across a landscape. Typically offered at lower-than-market interest rates, revolving loan funds are self-replenishing pools of money that utilize principal and interest payments on existing loans to issue new loans. They have been used effectively on small to large scale to develop businesses, assist healthcare, and improve environmental outcomes. Revolving loan funds also provide much-needed upfront capital for project startups. They are flexible and can be used in conjunction with more conventional funds, such as grants and loans.

For example, through a coalition of public and private partners, the Southwest Wildfire Impact Fund intends to utilize resources from private investors and revenues from biomass generated from forest thinning to offset the financial burden for wildfire mitigation in the San Juan National Forest wildland-urban interface. The project fosters regional collaboration through shared project financing and implementation. It also creates the opportunity for scaling up forest treatments and fire reduction by creating a revolving loan fund that

Table 2: Three-year hypothetical JPA budget showing revenue and expenses.

ITEM	YR1	YR2	YR3	TOTAL
REVENUE				
Contributions	60,000	60,000	60,000	180,000
Grants	60,000	120,000	150,000	330,000
Fee-for-Service	140,000	220,000	330,000	690,000
TOTAL REV-ENUE	260,000	400,000	540,000	1,200,000
EXPENSES				
Labor	340,000	370,000	410,000	1,120,000
Operations	20,000	30,000	40,000	90,000
Outreach	2,000	3,000	4,000	9,000
TOTAL EX-PENSES	362,000	403,000	454,000	1,219,000

reinvests proceeds into additional projects, ensuring that capital is available for long-term re-treatment and expansion of forest health interventions.

Blue Forest is operating a revolving loan fund called the California Wildfire Innovation Fund, which could expand into broader support for other forest health entities.

Parametric Insurance Parametric or index-based insurance covers the probability of a predefined event instead of indemnifying actual loss incurred (?). These so-called trigger events are typically disaster-related (e.g., wildfires, flooding, hurricanes, earthquakes) and are measured through triggers such as wind speed, quake magnitude, acres burned or burn severity, or rainfall amount. Insurable triggers are modeled and must happen by chance. When the triggers are reached, a predetermined pay-out is made regardless of the sustained physical losses. Parametric insurance is meant to complement existing indemnity insurance but is increasingly used for post-disaster restoration funding in the natural world.

One of the earliest examples of parametric insurance used for nature recovery is the Mesoamerican Reef parametric insurance that provided \$800,000 for reef restoration following Hurricane Delta. The trigger was windspeed with a parameter greater than 100 knots. The funds came from the [Coastal Zone Management Trust](#). Using insurance to protect forests and communities, likely through wildfire risk mitigation, could be a novel way to fund activities throughout the region and provide complementary funds that could be particularly valuable post-fire.

Tech-based Solutions The technology could help to connect funders to a new JPA. Blockchain and digital solutions have mostly been applied to reforestation and carbon sequestration projects. It may be possible to establish a digital marketplace where funders and implementers can connect and collaborate on implementing forest health restoration and infrastructure projects. Developing a platform for aggregated funding could be part of a funded project or developed in tandem rather than as a standalone initiative.

3 Budget

CAL FRAME partners estimated an approximate annual budget averaging \$400,000 during the 1st three years of operation and based on similar operating expenses for RCDs in the region (Table 2).

3.1 Revenue

Revenue contributions include contributions, grants, and fee-for-service consulting. Contributions and gifts may come from local to regional foundations and corporations interested in forest health. Individual contributions

always have the potential to add up to more than foundation and corporate gifts, but they require more time to manage. Creating a time-bound campaign with a specific fundraising goal replete with a thermometer to show progress could be a great way to involve communities in the region through giving and creating outreach or communications opportunities at the same time to explain the need for a JPA and the importance of sustainable funding from the community to protect homes, infrastructure, and forests.

In the case of pilot projects supported by the Office of Land Use and Climate Innovation, a tranche of endowment-like funding is being provided for startup funding of aggregation entities. In some cases, this amount is as much as \$1 million, which, along with a capital campaign to bolster individual giving and create awareness around a JPA, could create a base endowment and provide stable operational revenue to a new organization from the principal if the endowment funds are wisely invested.

Although not competing for grants with RCDs, a JPA may help administer a large grant across multiple RCDs to leverage more funds across a region. With the devolution of some state funding sources, this could be a great option for managing those funds and reducing competition for funding resources, as the grants are allocated to local organizations. The fee-for-service section includes an item for grant administration. It also includes an option for implementing landowner forest plans, as this may be a viable revenue source and could help source wood. A sort yard managed by the JPA to source, centralize, and sell woody biomass is another revenue option but will require investment to be successful. Expenses for the sort yard are included in the expenses section.

JPA - RCD competition

Evaluating whether to form a new governmental entity or organization is an important consideration for feedstock aggregation. Working with existing organizations, such as Resource Conservation Districts or Fire Safe Councils, would be a good starting point. It is critical to incorporate into the JPA's bylaws and revenue-generating practices that grant fundraising does not compete with RCDs. It is possible a JPA may collaborate on a grant with one or more RCDs but should never submit grants for which an RCD qualifies.

Contingency funding should be written into the budget expenses. Adding 10-15% contingency line items to any secured grant would supplement that funding, although most grant funding contingencies are to be applied to implementation budget shortfalls. Other contingency funding sources could include unrestricted funding (such as contributions or gifts) and a higher indirect rate.

3.2 Expenses

An annual expense budget of ~\$ \$400,000 is estimated for a JPA startup. The total expenses/year slowly ramp up each year of the budget with the idea that with additional secured revenue, a JPA would bring on more staff capacity and increase offerings, reach, or fee-for-service activities such as additional sort yards. The bulk of the expenses is for labor and staff, including an executive director, a contracted feedstock manager, an administrative bookkeeper, and various contracted services. Briefly, the staff responsibilities are the following:

- **Executive Director.** Manage board of directors, lead program development/outreach/education/fundraising/communications, develop an annual budget, manage contracts, recruit and manage staff.
- **Administrative Bookkeeper.** Track/invoice budget, develop cost allocation plan, manage grant reporting, administer payroll, manage AP/AR accounts, prepare 1099s and tax docs, develop grant budgets, and ensure financial compliance.
- **Feedstock Manager.** Source and sell feedstock from public and private lands, arrange transport to sort yard(s), manage sort yards, develop outreach materials, manage compliance documents for sort yard and aggregation, and maintain records.

Other expenses include operations and maintenance, audit and legal fees, insurance, equipment, software, travel, and bank fees. Expenses include communications, website development, outreach, equipment, insurance, and land lease costs for a sort yard.

Table 3: Timeline for a theoretical JPA startup period.

Task	Sub-task	2025	2026	2027	2028	2029	2030
1. Revenue Generation	Endowment creation			X	X	X	
	Gift campaign	X	X	X			
	Fee-service			X	X	X	X
	Sort yard 1					X	
	Sort yard 2						X
2. Staffing	Executive director	X					
	Administrator	X					
	Feedstock manager		X				
3. Governance	JPA bylaws	X	X				
	LAFCO filing	X					

3.3 Timeline

An approximate timeline for key activities to fund and operationalize a JPA is shown in Table 3. Endowment seed funding and a capital campaign will be critical to initiate the endowment. Staffing is described under expenses and shows the approximate start time for each staff member. The development of JPA bylaws will define governance, and JPA will be created when the Local Agency Formation Commission (LAFCO) filing is completed.

4 Recommendations

Funding JPAs is not a simple task, as most public funding sources are focused on project implementation, e.g., funding restoration projects in the field, rather than providing administrative support or organizational startup. Some foundations fund this type of work, but accessing those funds can be challenging. Another challenge is competing with existing entities for scarce local, regional, and federal resources. The following are recommendations to consider when examining the feasibility of building and maintaining a JPA for feedstock aggregation:

- 1. Utilize public funds for startup and traditional projects.** Public funding is readily available and generally suitable for trials or the initial development of feasible but new ideas.
- 2. Pilot non-traditional, higher risk, or new revenue sources as pilots.** Then, ramp them up with success, and participating entities and partners see their value and understand how they work.
- 3. Co-house staff and resources.** Utilize existing offices and share staff and other resources as capacity and funding allows. This is particularly important during the startup and early phases of developing a JPA.
- 4. Incorporate feedstock insurance.** To de-risk feedstock pricing and also attract new investors to wood product businesses.
- 5. Leverage public funds with private investment.** When public funds are secured, immediately work to leverage them with other public and private funding resources. Don't wait until near the end of grants to look for additional funds.
- 6. Sort yards with wood campuses or easements.** Could land trusts or agencies that fund easements include sort yards as part of restoration efforts?

7. **Use the right revenue source.** Property and sales tax increases are more effective in populated areas with higher incomes; however, they usually are not appropriate for disadvantaged rural where the tax base and population tend to be too low to provide enough funding.